Shariah-Coherent Blended Finance: A Model of Zakat–Waqf–Sukuk Credit Enhancement for Real-Economy Development

Abstract

Islamic finance holds, in waqf and zakat, two of the world's largest pools of social capital, and in sukuk a market instrument tied to real assets, yet no formal model blends the three to finance real-economy development. This paper develops a tractable single-factor structured-credit model of a facility in which zakat is transferred in ownership to eligible beneficiaries as first-loss equity, waqf income provides a subordinated charitable backstop, and a senior development-sukuk tranche raises private capital that the social layers de-risk. The central result formalises *Shariah-coherent credit enhancement* as a checkable structural condition: when each layer bears loss in proportion to its own capital, the senior tranche retains strictly positive tail risk, and no fee is paid for protection, the enhancement is feasible if and only if the first-loss capital is third-party donative (*tabarru*) subordination rather than a guarantee. The structure therefore reconciles credit enhancement with the loss-sharing principle (*al-ghunm bil-ghurm*) and avoids *riba* and *daman bi-ajr*. The model yields a closed-form mobilisation multiplier that is decreasing in default risk, asset correlation, and loss given default; a second, liquidity channel through which a waqf facility compresses the illiquidity premium that thin sukuk markets impose, so the structure mobilises through price as well as subordination; an incentive-compatibility condition under which de-risking does not slacken project screening; and a welfare condition under which owned equity dominates a one-time cash grant for the beneficiary. A calibration to Islamic-microfinance risk parameters illustrates an investment-grade senior tranche supporting roughly ten times its social capital. The contribution is conceptual and model-conditional: the calibration is not observed on a live portfolio, the juristic questions are placed before scholarly ruling, and a pilot constitutes the research agenda.

Keywords: Islamic social finance; blended finance; sukuk; zakat; waqf; structured credit; risk-sharing.

1 Introduction

Two facts sit uneasily together in Islamic finance. Annual zakat is estimated in the hundreds of billions of dollars and waqf assets in the trillions, yet much zakat is distributed as one-time consumption relief and much waqf is idle or under-deployed [Kahf, 1999, Obaidullah and Khan, 2008]. At the same time, the development finance the objectives of the Shariah most favour, for small enterprises, affordable housing, and sustainable infrastructure, is underserved, and the sukuk that could serve it trade in thin, illiquid secondary markets. Prior empirical work indicates that the binding constraint on the sukuk market is its thinness rather than any pricing property. Sukuk are significantly less liquid than matched conventional bonds, a gap most pronounced for corporate issuance and narrowing as the market matures [Almaskati, 2023]; in the author's companion study, secondary-market illiquidity in tradable sukuk is an order of magnitude or more above that of conventional bond funds by the Amihud measure [Amihud, 2002], and at longer horizons sukuk returns behave much like conventional asset-backed debt. The remedy implied by that diagnosis is not to seek a distinct return but to crowd primary capital into real-asset issuance.

Blended finance, in which concessional or philanthropic capital absorbs first loss to mobilise private investment, is the conventional instrument for exactly this problem [OECD, 2018]. Its application within Islamic finance, however, faces a constraint that conventional blended finance does not. Subordinating one layer to protect another can reconstitute an impermissible guarantee for a fee (*daman bi-ajr*), and unequal loss-bearing can offend the principle that entitlement to gain is coupled to the bearing of loss (*al-ghunm bil-ghurm*). Whether credit enhancement can be made compatible with these requirements is an open question, and it is the question this paper formalises.

The contribution is a tractable model of a three-layer facility and four results. First, a closed-form mobilisation multiplier links the private capital raised to the social capital committed, and signs its comparative statics in default risk, asset correlation, and loss given default; a complementary liquidity channel shows that a waqf liquidity facility compresses the illiquidity premium thin secondary markets impose, so the facility mobilises through price as well as subordination, directly addressing the market-thinness constraint that motivates the paper. Second, a characterisation *formalises* a settled juristic position as a checkable structural condition: credit enhancement that preserves loss-follows-capital, leaves the senior tranche with strictly positive tail risk, and charges no fee for protection is feasible if and only if the first-loss capital is third-party donative subordination. The result converts the requirement into model terms rather than discovering it; its value is to make Shariah-coherence verifiable from a capital structure. Third, an incentive-compatibility condition shows when de-risking does not slacken the screening of projects, addressing the standard moral-hazard objection to concessional capital. Fourth, a welfare condition identifies when transferring zakat as owned, at-risk equity dominates a one-time cash grant for the recipient.

The paper is explicit about its register. It is a conceptual contribution: its quantitative statements are conditional on the model and a calibration to comparable evidence, not observed on a live portfolio; the juristic questions it raises are placed before a Shariah board for ruling rather than asserted as settled (Appendix B); and a capped pilot, not a further regression, is the validation it proposes. Section 2 reviews the literature; Section 3 sets out the institutional and juristic framework and the formal model; Section 4 presents the analytical and numerical findings; Section 5 discusses implications and limitations; Section 6 concludes.

2 Literature review

The paper joins five literatures that have remained largely separate.

Blended finance and first-loss capital. Blended finance, the use of concessional or philanthropic capital to absorb first loss and mobilise private investment, has become a central instrument for closing financing gaps in emerging economies [OECD, 2018, Havemann et al., 2020], from climate-resilient infrastructure [Delina et al., 2026] to SMEs [Khan and Badjie, 2022]. Its record is mixed: public–private and blended models frequently fall short of mobilisation targets [Chowdhury and Tadjoeddin, 2023], reliance on private development finance carries long-run policy risks [Preston, 2026], and project success hinges on country and structural factors [Eberhard and Gratwick, 2011]. The recurring danger is that subsidised risk-bearing distorts screening and crowds out market discipline rather than crowding capital in: evidence on development-finance institutions documents such unintended outcomes [Léon, 2025], philanthropic "de-risking" has been criticised for shifting risk onto concessional balance sheets [Sklair and Gilbert, 2022], and the security-design literature formalises the remedy as an originator-retention requirement under moral hazard [Malekan and Dionne, 2014]. This paper imports the mobilisation mechanism and confronts its moral-hazard problem directly (Proposition 4), adding a constraint absent from this literature: in Islamic finance the first-loss subordination cannot be a guarantee for a fee and must be permissible under Shariah contract law.

Islamic social finance: zakat and waqf. Islamic social finance rests on zakat and waqf, large pools of social capital long restricted to consumption relief but increasingly argued to belong in productive deployment [Kahf, 1999, Obaidullah and Khan, 2008]. Directing zakat into productive capital strengthens beneficiary independence [Hariyanto et al., 2020] and, with digital administration [Komarudin et al., 2023] and an understanding of donor behaviour [Hakimi et al., 2021], raises both reach and welfare [Zauro et al., 2020, Widiastuti et al., 2021]; benevolent-loan models even shield the poor from predatory lending [Hanifuddin et al., 2024], though multidimensional poverty among the *asnaf* remains the target [Zailani et al., 2025]. Waqf, historically a patient charitable endowment [Kasdi et al., 2022], is now tied to development and environmental stewardship [Mangunjaya, 2023, Zawawi et al., 2023] and proposed as the complement zakat alone cannot supply [Atan and Johari, 2017], mobilised through real-estate investment trusts [Hasan and Sulaiman, 2016], crowdfunding and ecosystem models [Khan et al., 2023], takaful-style aid funds [Zakaria et al., 2019], and cash-waqf cooperatives [Zabri and Mohammed, 2018], while its governance demands institutional capacity rather than mere formal control [Shamim, 2026]. Existing cash-waqf-linked structures, however, park waqf in low-risk paper for charity coupons and mobilise no private capital [Qadri et al., 2024]. Read together, this strand establishes that zakat and waqf can serve as productive, welfare-improving development capital rather than mere relief, which is the premise on which the model's social layers rest; the present model differs by transferring ownership to the beneficiaries and treating the loss-sharing constraint formally.

Structured credit, sukuk, and securitisation risk. The private layer is mobilised through sukuk, whose market has grown fast enough to invite bibliometric and empirical scrutiny [Athief et al., 2025, Uluyol, 2023] and whose development hinges on depth and liquidity [Smaoui and Khawaja, 2017]. The model's machinery is the single-factor portfolio-credit framework that underlies modern capital regulation [Merton, 1974, Vasicek, 2002, Gordy, 2003], whose Islamic application is nascent [Abdul Halim et al., 2020]. A central concern is whether sukuk genuinely share risk or replicate debt: evidence of risk-shifting in debt-based sukuk [Hamzah et al., 2018] echoes the securitisation features blamed for the global financial crisis [Ayoub, 2012]. The juristic hurdle is the allocation of ownership risk (*daman al-milkiyyah*), whose misallocation yields formally compliant but substantively deficient products [Abdul Razak and Saupi, 2017, Asyiqin et al., 2025], and asset-based sukuk draw concern where investors lack genuine ownership [AbdulKareem and Mahmud, 2024]. Because a guarantor cannot charge a fee [Busari et al., 2024, Issoufou and Oseni, 2015], credit enhancement cannot be a paid guarantee; and the realism of senior risk is underscored by work on sukuk default [Zaheer and van Wijnbergen, 2025]. This motivates the requirement (Proposition 3) that the senior claim retain genuine tail risk.

The Islamic moral economy and risk-sharing. The constraints trace to the Islamic moral economy, which prizes genuine risk-sharing over fixed-return debt [Asutay, 2012, Askari et al., 2012], yet whose practice often drifts toward conventional benchmarks [Korkut and Ozgur, 2017, Tarriko and Hakmaoui, 2023]. Its central tenet, *al-ghunm bil-ghurm*, couples entitlement to gain with the bearing of loss, so unequal loss-bearing among co-venturers is impermissible unless structured outside the partnership [Abdul Razak and Saupi, 2017]. Participatory contracts that grant the poor ownership of real assets operationalise this principle [Islam and Ahmad, 2022] but require strict governance to prevent failure [Naim et al., 2016] and value-creating dispute resolution [Corgnet et al., 2025]; donative contracts (*tabarru*) supply the permissible route to third-party support [Atmeh and Maali, 2017, Al-Ayyubi et al., 2023]. The general finance literature, in turn, shows that genuine risk-sharing carries measurable stabilising value [Weagley, 2019]. The facility must therefore reconcile credit enhancement with risk-sharing, the problem Proposition 3 resolves.

Mitigating moral hazard in Islamic microfinance. Risk-sharing meets its hardest test in microfinance, where information asymmetry and strategic default are acute and roughly a fifth of the Muslim poor decline interest-based credit [Cameron et al., 2021]. Introducing concessional first-loss capital invites the standard objection that it dulls screening; the documented remedies are peer monitoring, group lending, and retained originator stakes [Cameron et al., 2021, Fan et al., 2019], set against the structural challenges of financing the poor [Wulandari and Kassim, 2016]. Islamic microfinance nonetheless shows distinctive performance [Ashraf et al., 2014], crisis resilience [Ben Salem et al., 2023], and a positive link between prudent credit-risk-taking and efficiency [Fersi and Boujelbene, 2023, Mohamed and Elgammal, 2023]. These findings calibrate the model and motivate its incentive-compatibility condition (Proposition 4): charitable capital absorbs loss only above a retained originator stake.

We are aware of no *formal model* that joins these strands. Proposals to deploy zakat and waqf as first-loss or blended development capital are emerging [Qadri et al., 2024, Zabri and Mohammed, 2018], and waqf-sukuk hybrids have been designed [Zain and Muhamad Sori, 2020], but they neither transfer ownership to the beneficiaries through *tamlik* nor formalise the loss-sharing constraint as a condition a capital structure can satisfy. That specific combination, ownership transfer plus a formal credit model that turns the juristic constraints into checkable structural conditions, is the paper's opening: the framework and model of Section 3 and the findings of Section 4 supply both.

3 Methodology

The methodology has two parts: the facility's contracts and the Shariah principles that constrain them (Section 3.1), and the formal credit model built on those constraints (Section 3.2).

3.1 Institutional and juristic framework

The facility finances a portfolio of vetted real-economy projects through three layered claims, each governed by its own contract.

- **Zakat as beneficiary grant-equity.** Zakat is transferred in full ownership (*tamlik*) to eligible recipients (the *asnaf*) as their equity in the projects. Being equity, it bears first loss by its nature. Whether an at-risk equity stake validly discharges the zakat obligation is a juristic question placed before a Shariah board, not settled here (Appendix B).
- **Waqf as patient backstop.** Waqf income (the corpus preserved) provides a capped, uncompensated subordinated backstop aligned to the waqf's own beneficiaries, the project communities.
- **Senior development sukuk.** Asset-backed, risk-sharing sukuk are sold to private investors, who bear genuine residual and tail risk.

Three principles operate as binding design constraints rather than as commentary, and the model treats them as such.

Assumption 1 (Loss follows capital). *Each layer bears loss only up to its own committed capital, and proportionally within the layer. This encodes the coupling of entitlement to gain with the bearing of loss,* al-ghunm bil-ghurm*, and the associated notion of ownership risk in Islamic contract [Abdul Razak and Saupi, 2017].*

Assumption 2 (No fee for protection). *No party receives a fee in exchange for absorbing another party's loss; first-loss support is a donation (*tabarru*) by a party that is neither a paid agent nor a co-venturing partner of the protected investor. This excludes a guarantee for a fee,* daman bi-ajr*, while remaining within the recognised treatment of donative contracts and third-party support in Islamic finance [Al-Ayyubi et al., 2023, Atmeh and Maali, 2017, Busari et al., 2024, Issoufou and Oseni, 2015].*

Assumption 3 (Genuine senior risk). *The senior tranche retains strictly positive probability of loss; its principal is not guaranteed. With asset backing and no purchase undertaking at par, this excludes* riba*; that senior risk is genuine rather than nominal is consistent with the securitisation-design and sukuk-default evidence [Abdul Halim et al., 2020, Zaheer and van Wijnbergen, 2025].*

The facility differs from Indonesia's Cash Waqf Linked Sukuk and related cash-waqf blended designs, which park waqf in government or low-risk sukuk and route the coupon to charity [Qadri et al., 2024, Zabri and Mohammed, 2018]: those structures mobilise no private capital, de-risk no instrument, and transfer no ownership. Here the social capital is put to work to crowd private capital in, and the poor own.

3.2 Model setup

A portfolio of N statistically identical projects is financed. Project i defaults when a latent asset variable falls below a threshold,

$$A_i = \sqrt{\rho}\, M + \sqrt{1-\rho}\, \varepsilon_i, \qquad M, \varepsilon_i \overset{\text{iid}}{\sim} N(0,1),$$

where M is a systematic factor, ε_i is idiosyncratic, and $\rho \in (0,1)$ is the asset correlation. Default occurs if $A_i < \Phi^{-1}(p)$, so the unconditional default probability is p. Loss given default is $\lambda \in (0,1]$. In the large-portfolio limit the realised loss rate conditional on the factor is, by the standard single-factor result [Vasicek, 2002, Gordy, 2003],

$$L(M) = \lambda\, g(M), \qquad g(M) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\, M}{\sqrt{1-\rho}}\right). \tag{1}$$

The senior development sukuk is the funded instrument, normalised to unit notional. The social layers form a first-loss buffer of size $a = z + w$ (zakat grant-equity z, waqf backstop w) subordinated beneath it, so the structure is over-collateralised by a. Writing L for the portfolio loss as a fraction of senior notional, the buffer absorbs $\min(L, a)$ and the senior tranche absorbs only the excess $\max(L - a, 0)$.

Definition 1 (Senior expected loss). *The expected loss per unit of senior notional is*

$$\mathrm{EL}_s(a) = \mathbb{E}\big[\max\left(\lambda\, g(M) - a,\, 0\right)\big]. \tag{2}$$

Financing depends not only on the senior tranche's credit quality but on the price investors pay for it. In a thin secondary market investors require an illiquidity premium over the credit-fair return, so the capital raised per unit of senior notional is the present value of its cashflows discounted at $r_f + c(a) + \pi$, where r_f is the risk-free rate, $c(a)$ is a credit spread that vanishes as $\mathrm{EL}_s(a) \to 0$, and $\pi \geq 0$ is the illiquidity premium. For an investment-grade tranche $c(a) \approx 0$, so financing is governed by $r_f + \pi$; write $A(r_f, T) = \big[1 - (1 + r_f)^{-T}\big]/r_f$ for the annuity factor at tenor T.

Assumption 4 (Thin secondary market). *The senior sukuk trades in a thin secondary market, so investors require a strictly positive illiquidity premium $\pi > 0$, consistent with the substantial illiquidity documented for sukuk [Almaskati, 2023] and quantified in the author's companion study. The waqf standing liquidity facility, funded from waqf income, together with investment-grade eligibility, compresses this premium.*

4 Findings

This section reports the model's analytical findings (Propositions 1–5), each proved in Appendix A, followed by a numerical illustration.

4.1 Mobilisation

Proposition 1 (Mobilisation multiplier). *Define the mobilisation multiplier* $m(a) = 1/a$*, the senior (private) capital raised per unit of social capital committed to the buffer. For any target senior expected loss* $\varepsilon^* \in (0, \mathrm{EL}_s(0))$ *there is a unique minimal buffer* a^* *solving* $\mathrm{EL}_s(a^*) = \varepsilon^*$*, and the implied multiplier is* $m^* = 1/a^*$*. Moreover* a^* *is increasing, and hence* m^* *decreasing, in the default probability* p*, the asset correlation* ρ*, and the loss given default* λ*.*

The result quantifies the trade-off at the centre of any blended structure: a thicker charitable buffer buys a safer, larger senior tranche, and riskier portfolios require more charity per unit mobilised. The multiplier is leverage on genuine charity, not a free lunch; the social layers bear in expectation what the senior tranche is spared.

4.2 Liquidity and the cost of capital

Credit subordination is only one of the facility's two financing channels. The second works through the price of the senior claim, and it is the channel that speaks to the market-thinness constraint motivating the paper.

Proposition 2 (Liquidity channel). *Hold credit quality fixed at investment grade,* $c(a) \approx 0$*. A waqf liquidity facility that compresses the illiquidity premium by* $\ell \in (0, \pi]$ *raises the capital mobilised per unit of senior notional by* $\ell\, A(r_f, T)$*, where* $A(r_f, T) = [1 - (1 + r_f)^{-T}]/r_f$ *is the annuity factor at tenor* T*. The facility therefore mobilises through two compounding channels, credit subordination (Proposition 1) and illiquidity compression, of which a credit-only blended structure exploits only the first.*

The liquidity gain is a financing transfer, not a creation of value: the waqf bears, on behalf of its own beneficiaries, the illiquidity it removes from investors. Numerically (Section 4.6), compressing a premium of the order measured for tradable sukuk is worth several percent of senior notional in present value, comparable to the buffer the social layers contribute, so the two channels are of similar economic weight. This is the methodological response to the objection that a credit model is silent on the illiquidity it is meant to address.

4.3 Shariah-coherent de-risking

The next result is the paper's core. It characterises exactly which credit enhancements are admissible under the framework of Section 3.1.

Definition 2 (Admissible enhancement). *A* credit enhancement *is any rule that lowers* EL_s *by assigning the absorbed loss to a bearer. The mechanism space* $\mathcal{E}$ *is enumerated by bearer and consideration: (i) reallocation of loss among the senior investors and their co-venturing partners; (ii) a third party's commitment to cover senior losses for a fee; (iii) a third party's uncompensated, subordinated donation. An enhancement is* admissible *if it satisfies Assumptions 1–3.*

Proposition 3 (Characterisation of Shariah-coherent credit enhancement). *Consider any arrangement that reduces the senior tranche's expected loss from its unenhanced level* $\mathrm{EL}_s(0)$ *to* $\mathrm{EL}_s(a) < \mathrm{EL}_s(0)$*. Under Assumptions 1–3, the enhancement is feasible if and only if the reduction is financed by third-party donative (*tabarru*) subordinated capital, of size* $a = z + w$*, that bears the first loss* $\min(L, a)$*. Equivalently: (i) no intra-partner reallocation of loss and no fee-bearing guarantee can achieve the reduction without violating Assumption 1 or 2; and (ii) donative subordination achieves it while leaving the senior tranche with strictly positive tail risk,* $\Pr[L > a] > 0$*, as required by Assumption 3.*

Proposition 3 is the sense in which the structure is Shariah-coherent. Credit enhancement is usually delivered by a guarantee, which Assumption 2 forbids, or by reallocating losses among co-venturers, which Assumption 1 forbids once it breaks proportionality to capital. The only remaining route is a donor, outside the investment partnership and unpaid, whose capital is genuinely subordinated. The zakat equity satisfies this as the property of the *asnaf*, who bear first loss as owners; the waqf backstop satisfies it as an uncompensated donation aligned to its own beneficiaries. The senior investors are not guaranteed, since they retain the tail. The result thus turns a juristic requirement into a structural one: in Islamic finance, permissible credit enhancement *is* third-party donative subordination. It is a selection over the mechanism space $\mathcal{E}$ of Definition 2, not a tautology: its content is that the three independently motivated juristic constraints are jointly satisfiable, and by a unique element of $\mathcal{E}$, one that zakat and waqf render implementable at scale. The economic substance lies in that implementability, which Sections 4 and 5 develop.

4.4 Incentives

Concessional first loss invites the standard objection that it dulls the originator's incentive to screen. Let the originator choose screening effort $e \geq 0$ at cost $\psi(e)$, with $\psi' > 0, \psi'' > 0$, which lowers the default probability, $p'(e) < 0$. If the originator bears none of the loss, it sets $e = 0$.

Proposition 4 (Incentive compatibility). *Suppose the originator retains a fraction $\beta \in [0, 1]$ of the first-loss equity (or, equivalently, a performance fee linked to portfolio survival). Then positive screening effort $e^* > 0$ is incentive-compatible if and only if $\beta \geq \beta^*$, where β^* is increasing in the marginal cost of effort and decreasing in the sensitivity of portfolio loss to effort. For $\beta \geq \beta^*$ the de-risking does not slacken screening.*

The condition makes the governance requirement precise: charity may absorb loss only above a retained skin-in-the-game stake, so that the party choosing projects still internalises their quality. It is the formal answer to the moral-hazard critique of blended finance.

4.5 Beneficiary welfare

Finally, transferring zakat as at-risk equity rather than as cash is justified only if it serves the recipient. Let a beneficiary with CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$ and baseline wealth W_0 choose between a one-time cash grant g and an equity stake of the same initial value that returns $R > 1$ if the project survives, with survival probability σ, and a recovery $\kappa < 1$ otherwise.

Proposition 5 (When ownership dominates a grant). *There is a threshold γ^*, increasing in the survival probability σ and the upside R, such that the equity stake yields higher expected utility than the cash grant if and only if the beneficiary's risk aversion satisfies $\gamma < \gamma^*$. For projects that survive in the large majority of states and offer meaningful upside, γ^* is high, so ownership dominates for all but the most risk-averse recipients.*

Proposition 5 states honestly what the structure asks of the poor: an owned, appreciating, but risky asset in place of certain consumption. It dominates when projects are sound and upside is real, and it is a condition placed before the Shariah board, not assumed (Appendix B).

Proof sketches for Propositions 1–5 are collected in Appendix A.

4.6 Numerical illustration

The propositions are signed analytically; a calibration shows the magnitudes are economically meaningful. The default probability is anchored to Islamic-microfinance evidence, where global portfolio-at-risk runs at roughly four to five percent [Mohamed and Elgammal, 2023, Fersi and Boujelbene, 2023, Fan et al., 2019, Obaidullah and Khan, 2008]; we set $p = 4.5\%$, loss given default $\lambda = 45\%$, asset correlation $\rho = 0.15$, tenor $T = 5$ years, and an illiquidity premium

Table 1: Calibrated illustration of the facility. Figures are model-conditional, not observed on a live portfolio.

Quantity	Value
Portfolio expected loss	$\approx 2.0\%$
First-loss attachment $a = z + w$ (zakat 6%, waqf 4%)	10%
Senior tranche expected loss EL_s	≈ 1.2 bps (investment grade)
Probability the senior tranche takes any loss	$\approx 0.6\%$
Senior 99.5% tail loss (finite pool)	≈ 35 bps
Mobilisation multiplier $m = 1/a$	$10\times$
Zakat development-multiplier vs a one-time grant	$\approx 16.7\times$
Illiquidity premium compressed by the waqf facility	100 bps
PV financing gain, liquidity channel $\ell\, A(r_f, T)$	$\approx 4.3\%$ of senior notional

$\pi = 100$ bps, of the order indicated by the companion study's illiquidity evidence [Almaskati, 2023], on a portfolio of $N = 300$ projects with a first-loss buffer $a = 10\%$ (zakat 6%, waqf 4%).

The senior tranche is investment grade by expected loss, and the facility's two financing channels are of comparable economic weight: the social buffer is 10% of senior notional, while compressing the illiquidity premium is worth about 4.3% in present value (Proposition 2). The reported figures are the finite-pool ($N = 300$) Monte Carlo, whose 99.5% senior tail of about 35 bps exceeds the large-portfolio (asymptotic) value of roughly 11 bps by the idiosyncratic granularity the ASRF limit suppresses. The sensitivity surface (Table 2) traces the senior expected loss as the pool deteriorates.

Table 2: Senior expected loss (basis points) by default probability p and asset correlation ρ, at a 10% first-loss attachment. Source: companion simulation, `wzs_sensitivity.csv`.

$p \setminus \rho$	0.10	0.15	0.25
5%	0.3	1.8	8.6
8%	3.3	9.6	27.9
12%	19.0	36.1	73.1

The full model, parameters, and sensitivity analysis are reproducible from the accompanying open code.[1]

4.7 Robustness

Three considerations bound the calibration. First, the sensitivity surface (Table 2) shows the senior tranche stays investment grade for default rates to 8% at correlations to 0.15, and that the 10% buffer is materially eroded only under jointly severe stress ($p = 12\%$, $\rho = 0.25$); higher-risk pools require a thicker buffer, exactly as Proposition 1 predicts. Second, the Gaussian single-factor structure understates tail dependence; a heavier-tailed or multi-factor specification would raise the required attachment a^* but leave the signed comparative statics, and hence the two-channel mobilisation result, intact. Third, holding loss given default fixed is conservative for the mobilisation claim: stochastic, correlated recovery would widen the senior tail and is absorbed by the same buffer-sizing logic. None of these overturns the results; each scales the buffer.

[1] Simulation and sensitivity outputs are produced by `src/innovation/wzs_simulation.py` and reported in `docs/results/wzs_sensitivity.csv` in the companion repository.

5 Discussion

Three implications follow. First, the mobilisation result reframes the social capital not as a subsidy consumed but as a catalyst: a modest, genuinely charitable buffer crowds a multiple of private capital into real-economy issuance, directly addressing the market-thinness constraint that motivates the paper. The honest accounting is that the leverage is borne by charity in expectation; the structure is defensible only because the donors, the *asnaf* as owners and the waqf for its own beneficiaries, are also the intended development targets.

Second, Proposition 3 clarifies a point that travels beyond this instrument: the Islamic prohibition on guarantees-for-fee and the loss-sharing principle do not preclude credit enhancement but pin down its form. The result formalises, rather than discovers, the admissibility of unpaid third-party support; its contribution is to convert that settled juristic position into a structural condition checkable from the capital stack, offering the moral-economy literature a concrete reconciliation of risk-sharing with market mobilisation [Asutay, 2012].

Third, Proposition 4 answers the moral-hazard objection on its own terms. Blended finance is rightly criticised when concessional capital removes the originator's incentive to screen [Léon, 2025, Malekan and Dionne, 2014]; here the retained-stake condition restores it, in the spirit of the monitoring and group-incentive mechanisms documented for Islamic microfinance [Cameron et al., 2021, Fan et al., 2019], and the governance design is not optional decoration but a model requirement. For financial stability, the facility's promise is depth: by both de-risking issuance and supplying liquidity (Proposition 2), it deepens a thin sukuk market on both margins rather than waiting for liquidity to arrive.

5.1 Limitations and research agenda

The paper is conceptual, and its limits are the conditions of that genre. The calibration is to comparable evidence, not observed on a live portfolio; the figures in Section 4.6 are model-conditional and should be read as illustrations of the analytics, not as findings. The single-factor Gaussian structure, exogenous default probability, and identical-projects assumption are simplifications whose relaxation is left to future work. The juristic questions, in particular whether zakat may take the form of at-risk equity and whether the discharge of the obligation is valid, are open and are placed before a Shariah board in Appendix B rather than resolved here. External validity to any specific jurisdiction is an extrapolation: the model is general, a national application is illustrative, and a capped pilot is the proper test. The research agenda is therefore threefold: a scholarly ruling on the istifta; a pilot generating the first live data; and, with that data, the empirical estimation the present design only motivates.

6 Conclusion

Islamic finance is often urged to live up to its risk-sharing, real-economy promise, and as often told that its prohibitions make instruments like credit enhancement impossible. This paper shows the two can be reconciled. A facility that transfers zakat to the poor as first-loss equity, deploys waqf income as a patient charitable backstop, and sells a senior development sukuk to private investors mobilises private capital for the real economy while preserving loss-follows-capital, avoiding a guarantee for a fee, and leaving genuine risk with the investor. The mobilisation it achieves, the incentive condition it must satisfy, and the welfare trade-off it asks of the poor are all stated precisely, and its juristic conditions are placed before scholarly ruling rather than assumed. The structure turns a diagnosed problem, the thinness of Islamic capital markets, into a mechanism that crowds capital in.

A Proofs

Proof of Proposition 1. $\mathrm{EL}_s(a) = \mathbb{E}\big[(\lambda g(M) - a)^+\big]$ is continuous and strictly decreasing on $[0, \lambda)$, since

$$\frac{d}{da}\,\mathbb{E}\big[(\lambda g(M) - a)^+\big] = -\Pr[\lambda g(M) > a] < 0 \qquad (a < \lambda),$$

falling from $\mathrm{EL}_s(0) = \mathbb{E}[\lambda g(M)]$ to 0 at $a = \lambda$. Continuity and strict monotonicity yield a unique a^* with $\mathrm{EL}_s(a^*) = \varepsilon^*$ for any $\varepsilon^* \in \big(0, \mathbb{E}[\lambda g(M)]\big)$, and $m^* = 1/a^*$ since $a \mapsto 1/a$ is strictly decreasing.

For the comparative statics, fix a. In (1), $\partial g/\partial p > 0$ pointwise (a higher p raises $\Phi^{-1}(p)$ and hence the conditional default rate at every M), so a larger p induces first-order stochastic dominance in $\lambda g(M)$; as $x \mapsto (x - a)^+$ is nondecreasing, $\mathrm{EL}_s(a)$ increases in p. A larger ρ raises $\mathrm{EL}_s(a)$ by the standard tail-monotonicity of the single-factor conditional-default distribution in the correlation parameter [Gordy, 2003], applied to the convex map $x \mapsto (x - a)^+$. Finally $\mathrm{EL}_s(a)$ is increasing in λ since $\lambda g(M)$ scales in λ. In each case $\mathrm{EL}_s(\cdot)$ shifts up pointwise, so the unique solution of $\mathrm{EL}_s(a^*) = \varepsilon^*$ rises, and $m^* = 1/a^*$ falls. □

Proof of Proposition 3. ($\Leftarrow$) Let the reduction from $\mathrm{EL}_s(0)$ to $\mathrm{EL}_s(a)$ be financed by subordinated capital of size $a = z + w$ contributed as a gift by parties outside the senior investment, bearing the first loss $\min(L, a)$. Each contributing layer loses at most its own capital and, within a layer, proportionally to it, so Assumption 1 holds. No consideration is paid for the loss absorption, so Assumption 2 holds. For $a < \lambda$, $\Pr[L > a] = \Pr[\lambda g(M) > a] > 0$ by (1), so the senior tranche retains strictly positive tail risk and Assumption 3 holds. The enhancement is therefore feasible.

($\Rightarrow$) Suppose a feasible enhancement reduces EL_s. Any reduction must transfer expected loss away from the senior tranche to some bearer. Three cases exhaust the alternatives to third-party donative subordination, and each is excluded. (i) If the loss is reallocated among the senior investors and their co-venturing partners other than in proportion to contributed capital, Assumption 1 is violated. (ii) If a party commits to absorb senior losses in exchange for any fee or compensated consideration, the arrangement is a guarantee for a fee (*daman bi-ajr*) and violates Assumption 2. (iii) If the arrangement removes the senior tail entirely (e.g. a principal guarantee), then $\Pr[L > a] = 0$ and Assumption 3 is violated. The only remaining bearer is capital that is (a) outside the senior investment partnership, (b) uncompensated, and (c) genuinely subordinated so that it, not the senior tranche, takes the first loss; this is precisely third-party *tabarru* subordination. Feasibility thus holds if and only if the enhancement takes that form. □

Proof of Proposition 2. The illiquidity premium raises the senior tranche's required yield by π, that is, an additional financing cost of π per unit of senior notional in each period. Compressing it by ℓ saves a cashflow of ℓ per period over the tenor T; discounted at the risk-free rate, the present value of that saving per unit of senior notional is $\ell \sum_{t=1}^{T} (1 + r_f)^{-t} = \ell\, A(r_f, T)$, with $A(r_f, T) = [1 - (1 + r_f)^{-T}]/r_f$. Because the waqf assumes the premium it removes, the gain is a transfer, not new value. □

Proof of Proposition 4. Let $\Lambda(e) = \mathbb{E}[\text{loss} \mid e]$ with $\Lambda'(e) < 0$ and the originator retaining a fraction β of the first-loss layer. The originator solves $\max_{e \ge 0} \{-\beta\,\Lambda(e) - \psi(e)\}$, with strictly convex cost ψ ($\psi' > 0, \psi'' > 0$). The objective is strictly concave in e, so the solution is characterised by the Kuhn–Tucker condition: an interior optimum $e^* > 0$ satisfies $-\beta\,\Lambda'(e^*) = \psi'(e^*)$, and exists iff the marginal benefit of effort at the origin exceeds its marginal cost, $-\beta\,\Lambda'(0) \ge \psi'(0)$; otherwise $e^* = 0$. Rearranging gives the threshold $\beta^* = \psi'(0)/\big(-\Lambda'(0)\big)$, so positive screening effort is incentive-compatible iff $\beta \ge \beta^*$. The threshold β^* is increasing in the marginal cost of effort $\psi'(0)$ and decreasing in the loss-sensitivity of effort $-\Lambda'(0)$. □

Proof of Proposition 5. Define $\Delta(\gamma) = \sigma\, u(W_0 + Rg) + (1-\sigma)\, u(W_0 + \kappa g) - u(W_0 + g)$ with $u(c) = c^{1-\gamma}/(1-\gamma)$, $R > 1 > \kappa \geq 0$. At $\gamma \to 0$, u is linear and $\Delta \to g\,[\sigma R + (1-\sigma)\kappa - 1]$, which is positive whenever the equity stake has nonnegative expected excess value, $\sigma R + (1-\sigma)\kappa > 1$. As $\gamma \to \infty$ the certainty-equivalent of the risky stake is governed by its worst outcome $W_0 + \kappa g < W_0 + g$, so $\Delta < 0$. Since u exhibits decreasing marginal utility, increasing γ penalises the mean-preserving dispersion of the stake relative to the certain grant, so Δ is strictly decreasing in γ. By continuity and strict monotonicity Δ has a unique zero γ^*, and the equity stake dominates iff $\gamma < \gamma^*$. Comparative statics: $\partial\Delta/\partial\sigma = u(W_0 + Rg) - u(W_0 + \kappa g) > 0$ and $\partial\Delta/\partial R > 0$, so the crossing point γ^* rises in both σ and R. □

B Juristic compliance: questions placed before scholarly ruling

The model's Assumptions 1–3 encode juristic requirements whose application to this structure is for a Shariah Supervisory Board to rule on, not for this paper to assert. The principal questions are: whether the transfer of zakat into a fully owned but initially illiquid, at-risk equity stake validly discharges the obligation under *tamlik*; whether a portion of zakat entitlement may take the form of first-loss equity consistent with the *maslaha* of the *asnaf*; whether the subordination of beneficiary equity and a third-party waqf *tabarru* backstop, with the senior tranche retaining genuine tail risk, remains free of an impermissible guarantee and consistent with *al-ghunm bil-ghurm*; whether an uncompensated, income-funded waqf backstop preserves perpetuity and qualifies as *tabarru* rather than *daman bi-ajr*; and whether the senior sukuk, asset-backed and without a par-value purchase undertaking, satisfy the prohibition of *riba*. Standards are referenced at the level of the relevant AAOIFI Shari'ah Standards (No. 17 Sukuk, No. 33 Waqf, No. 35 Zakah) and contemporary productive-zakat scholarship; precise rulings are sought before any pilot.

References

Zairihan Abdul Halim, Janice How, Peter Verhoeven, and M. Kabir Hassan. Asymmetric information and securitization design in islamic capital markets. *Pacific Basin Finance Journal*, 62, 2020. doi: 10.1016/j.pacfin.2019.101189.

Lutfi Abdul Razak and Muhammad Nabil Saupi. The concept and application of daman al-milkiyyah (ownership risk): Islamic law of contract perspective. *ISRA International Journal of Islamic Finance*, 9(2):148 – 163, 2017. doi: 10.1108/IJIF-06-2017-0002.

Ibraheem Alani AbdulKareem and Mohd Sadad Mahmud. Shariah issues of ownership and possession of asset-based sukuk structure: A shariah scholars' perspective. *UUM Journal of Legal Studies*, 15(1):171 – 196, 2024. doi: 10.32890/uumjls2024.15.1.8.

Solahuddin Al-Ayyubi, Evania Herindar, and Muhammad Nabhan Perdana. Maqasid sharia in tabarru' contract laws. *Jurnal Al-dustur*, 6(1):28 – 49, 2023. doi: 10.30863/aldustur.v6i1.4194.

Nawaf Almaskati. Revisiting the question of liquidity: are sukuk less liquid than conventional bonds? *Journal of Islamic Accounting and Business Research*, 14(7):1121–1132, 2023.

Yakov Amihud. Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56, 2002.

Ali Ashraf, M. Kabir Hassan, and William J. Hippler III. Performance of microfinance institutions in muslim countries. *Humanomics*, 30(2):162 – 182, 2014. doi: 10.1108/H-11-2013-0073.

Hossein Askari, Zamir Iqbal, Noureddine Krichene, and Abbas Mirakhor. *Risk Sharing in Finance: The Islamic Finance Alternative.* John Wiley & Sons, Singapore, 2012.

Mehmet Asutay. Conceptualising and locating the social failure of Islamic finance: aspirations of Islamic moral economy vs the realities of Islamic finance. *Asian and African Area Studies*, 11(2):93–113, 2012.

Istianah Zainal Asyiqin, M. Fabian Akbar, Dimas Putra Wirayudha, and Ayesha Binti Abul Hisyam. Reconciling ownership risk under islamic economic law: Comparative analysis of indonesian and malaysian contracts. *Indonesia Private Law Review*, 6(2):133 – 158, 2025. doi: 10.25041/iplr.v6i2.4545.

Nur Atika Binti Atan and Fuadah Binti Johari. A review on literature of waqf for poverty alleviation between 2006-2016. *Library Philosophy and Practice*, 2017(1), 2017.

Fauzul Hanif Noor Athief, Luthfiyyah Nurul Izzah, Darlin Rizki, and Afief El-Ashfahany. Intellectual structure of islamic capital market studies: A bibliometric approach. *Journal of Scientometric Research*, 14(1):221 – 238, 2025. doi: 10.5530/jscires.20250562.

Muhannad Ahmed Atmeh and Bassam Maali. An accounting perspective on the use of combined contracts and donations in islamic financial transactions. *Journal of Islamic Accounting and Business Research*, 8(1):54 – 69, 2017. doi: 10.1108/JIABR-07-2014-0024.

Sherif Ayoub. The global financial crisis, securitization and islamic finance: An opportunity for inward and outward reform. *ISRA International Journal of Islamic Finance*, 4(2):53 – 87, 2012. doi: 10.55188/ijif.v4i2.146.

Asma Ben Salem, Ines Ben Abdelkader, and Sameh Jouida. Is islamic microfinance a resilient business model during periods of crisis? empirical evidence from arab countries. *Palgrave CIBFR Studies in Islamic Finance*, Part F11442:205 – 238, 2023. doi: 10.1007/978-3-031-13302-2_10.

Sahed Busari, Hassan Suleiman, and Belal Salhab. Guarantee contracts in Islamic financial transaction: analysis of BNM resolution and AAOIFI standards. *Journal of Fatwa Management and Research*, 29(3):46 – 62, 2024. doi: 10.33102/jfatwa.vol29no3.585.

Alistair Cameron, Mandar Oak, and Yaping Shan. Peer monitoring and islamic microfinance. *Journal of Economic Behavior and Organization*, 184:337 – 358, 2021. doi: 10.1016/j.jebo.2021.02.001.

Anis Chowdhury and Mohammad Zulfan Tadjoeddin. Financing green transformation in developing countries. *Canadian Journal of Development Studies*, 44(3):430 – 453, 2023. doi: 10.1080/02255189.2022.2113767.

Brice Corgnet, Simon Gachter, and Roberto Hernan-Gonzalez. The contractual dispute resolution game: Real-effort experiments on contract negotiation and arbitration. *Journal of Economic Behavior and Organization*, 231, 2025. doi: 10.1016/j.jebo.2025.106902.

Laurence L. Delina, Madeleine Rachel Widia, Olivia Felice Janice Boedijanto, Sharon Feliza Ann P. Macagba, Johanne Rei R. Castro, and Rainbow Yi Hung Lam. Sequencing finance for climate resilience: Instruments, institutions and hong kong's role in mobilising private capital in southeast asia. *Business Strategy and the Environment*, 35(4):6089 – 6110, 2026. doi: 10.1002/bse.70473.

Anton Eberhard and Katharine Nawaal Gratwick. Ipps in sub-saharan africa: Determinants of success. *Energy Policy*, 39(9):5541 – 5549, 2011. doi: 10.1016/j.enpol.2011.05.004.

Yaoyao Fan, Kose John, Frank Hong Liu, and Luqyan Tamanni. Security design, incentives, and islamic microfinance: Cross country evidence. *Journal of International Financial Markets, Institutions and Money*, 62:264 – 280, 2019. doi: 10.1016/j.intfin.2019.08.002.

Marwa Fersi and Mouna Boujelbene. Financial and social efficiency analysis of islamic microfinance institutions. *International Journal of Emerging Markets*, 18(4):931 – 957, 2023. doi: 10.1108/IJOEM-02-2020-0197.

Michael B. Gordy. A risk-factor model foundation for ratings-based bank capital rules. *Journal of Financial Intermediation*, 12(3):199–232, 2003.

Fifi Hakimi, Tika Widiastuti, Muhammad Ubaidillah Al-Mustofa, and Rizqiyyah Al Husanaa'. Positive effect of attitude, peer influence, and knowledge zakat on zakat compliance behavior: Update in covid 19. *Journal of Islamic Economic Laws*, 4(2):1 – 16, 2021. doi: 10.23917/jisel.v4i2.13859.

Siti Raihana Hamzah, Obiyathulla Ismath Bacha, Abbas Mirakhor, and Nurhafiza Abdul Kader Malim. Empirical evidence of risk shifting in bonds and debt-based sukuk: The case of malaysian corporations. *Journal of Islamic Accounting and Business Research*, 9(5):687 – 700, 2018. doi: 10.1108/JIABR-06-2016-0068.

Iza Hanifuddin, Nur Kasanah, and Eficandra. Al-qard al-hasan program of bankziska: Zakat fund-based empowerment model for victims of loan sharks. *Juris: Jurnal Ilmiah Syariah*, 23 (1):1 – 12, 2024. doi: 10.31958/juris.v23i1.10799.

Erie Hariyanto, Muhammad Taufiq, Zainal Abidin, Miftahul Ulum, and Maimun. Effectiveness of the economic system to zakat and waqf for empowerment of the ummah in indonesia. *International Journal of Advanced Science and Technology*, 29(6):1910 – 1916, 2020.

Aznan Hasan and Syahnaz Sulaiman. The use of islamic real estate investment trust (i-reits) as a contemporary instrument in developing waqf assets: Potential structure, issues and challenges. *Intellectual Discourse*, 24:521 – 540, 2016.

Tanja Havemann, Christine Negra, and Fred Werneck. Blended finance for agriculture: exploring the constraints and possibilities of combining financial instruments for sustainable transitions. *Agriculture and Human Values*, 37(4):1281 – 1292, 2020. doi: 10.1007/s10460-020-10131-8.

Reazul Islam and Rubi Ahmad. Incorporation of mudarabah, musharakah and musharakah mutanaqisah with microfinance: A sustainable livelihood approach to poverty alleviation. *Journal of Economic Cooperation and Development*, 43(1), 2022.

Chaibou Issoufou and Umar A. Oseni. The application of third party guarantee in structuring sukuk in the islamic capital market: A preliminary literature survey. *Mediterranean Journal of Social Sciences*, 6(5):130 – 138, 2015. doi: 10.5901/mjss.2015.v6n5p130.

Monzer Kahf. The performance of the institution of zakah in theory and practice, 1999. International Conference on Islamic Economics towards the 21st Century, Kuala Lumpur.

Abdurrohman Kasdi, Abdul Karim, Umma Farida, and Miftahul Huda. Development of waqf in the middle east and its role in pioneering contemporary islamic civilization: A historical approach. *Journal of Islamic Thought and Civilization*, 12(1):186 – 198, 2022. doi: 10.32350/jitc.121.10.

Abidullah Khan, Raditya Sukmana, and Abid Mahmood. Islamic social finance ecosystem and the role of crowdfunding: A proposed model. *Hamdard Islamicus*, 46(3):91 – 111, 2023. doi: 10.57144/hi.v46i3.876.

Tariqullah Khan and Fatou Badjie. Islamic blended finance for circular economy impactful smes to achieve sdgs. *Singapore Economic Review*, 67(1):219 – 244, 2022. doi: 10.1142/S0217590820420060.

M. Komarudin, Ali Sophian, Hery Dian Septama, Titin Yulianti, Muhamad Ikhsan, and Tomy Pratama Zuhelmi. Ziswaf: Zakat application to improve ease of recording zakat data in indonesia. *WSEAS Transactions on Environment and Development*, 19:309 – 319, 2023. doi: 10.37394/232015.2023.19.27.

Cem Korkut and Onder Ozgur. Is there a link between profit share rate of participation banks and interest rate?: The case of turkey. *Journal of Economic Cooperation and Development*, 38(2):135 – 158, 2017.

Florian Léon. In the shadows of support: Unintended outcomes of dfis' intermediated lending in africa. *World Development*, 191:106998, 2025. doi: 10.1016/j.worlddev.2025.106998.

Sara Malekan and Georges Dionne. Securitization and optimal retention under moral hazard. *Journal of Mathematical Economics*, 55(1):74 – 85, 2014. doi: 10.1016/j.jmateco.2014.10.003.

Fachruddin M. Mangunjaya. How islamic tradition benefits nature and climate change action. *Journal of the Siam Society*, 111(2):253 – 264, 2023.

Robert C. Merton. On the pricing of corporate debt: the risk structure of interest rates. *The Journal of Finance*, 29(2):449–470, 1974.

Toka S. Mohamed and Mohammed M. Elgammal. Credit risk in islamic microfinance institutions: The role of women, groups, and rural borrowers. *Emerging Markets Review*, 54, 2023. doi: 10.1016/j.ememar.2022.100994.

Asmadi Mohamed Naim, Mahyuddin Abu Bakar, Muhammad Nasri Md. Hussein, and Mohamad Noor Habibi Long. Issues and challenges in offering mudarabah and musharakah products in islamic finance. *Jurnal Pengurusan*, 46, 2016.

Mohammed Obaidullah and Tariqullah Khan. Islamic microfinance development: Challenges and initiatives. Policy Dialogue Paper 2, Islamic Research and Training Institute (IRTI), Islamic Development Bank, 2008.

OECD. Making blended finance work for the sustainable development goals, 2018.

Daniel Preston. The long rise of private finance in development and the implications for policymakers. *Journal of Developing Societies*, 42(1):57 – 94, 2026. doi: 10.1177/0169796X251387437.

Resi Ariyasa Qadri, Raditya Hendra Pratama, Akhmad Khabibi, and Reza Syam Pratama. Refining the cash-waqf blended finance model for infrastructure development. *Management and Accounting Review*, 23(1):279–305, 2024.

Safaque Shamim. Historical evolution of waqf laws in india: From legislative engineering to governance in practice. *Legal Transformation in Muslim Societies*, 3(1):102 – 130, 2026.

Jessica Sklair and Paul Gilbert. Giving as "de-risking": Philanthropy, impact investment and the pandemic response. *Public Anthropologist*, 4(1):51 – 77, 2022. doi: 10.1163/25891715-bja10033.

Houcem Smaoui and Mohsin Khawaja. The determinants of sukuk market development. *Emerging Markets Finance and Trade*, 53(7):1501 – 1518, 2017. doi: 10.1080/1540496X.2016.1224175.

Dalila Tarriko and Abdelati Hakmaoui. Lack of profit-and-loss sharing contracts in moroccan islamic banks: An investigation during the coronavirus pandemic. *ISRA International Journal of Islamic Finance*, 15(3):160 – 178, 2023. doi: 10.55188/ijif.v15i3.616.

Burhan Uluyol. A comprehensive empirical and theoretical literature survey of islamic bonds (sukuk). *Journal of Sustainable Finance and Investment*, 13(3):1277 – 1299, 2023. doi: 10.1080/20430795.2021.1917224.

Oldrich Vasicek. The distribution of loan portfolio value. *Risk*, 15(12):160–162, 2002.

Daniel Weagley. Financial sector stress and risk sharing: Evidence from the weather derivatives market. *Review of Financial Studies*, 32(6):2456 – 2497, 2019. doi: 10.1093/rfs/hhy098.

Tika Widiastuti, Ilmiawan Auwalin, Lina Nugraha Rani, and Muhammad Ubaidillah Al Mustofa. A mediating effect of business growth on zakat empowerment program and mustahiq's welfare. *Cogent Business and Management*, 8(1), 2021. doi: 10.1080/23311975.2021.1882039.

Permata Wulandari and Salina Kassim. Issues and challenges in financing the poor: case of baitul maal wa tamwil in indonesia. *International Journal of Bank Marketing*, 34(2):216 – 234, 2016. doi: 10.1108/IJBM-01-2015-0007.

Mohd Zaidi Md Zabri and Mustafa Omar Mohammed. Examining the behavioral intention to participate in a cash waqf-financial cooperative-musharakah mutanaqisah home financing model. *Managerial Finance*, 44(6):809 – 829, 2018. doi: 10.1108/MF-05-2017-0189.

Sajjad Zaheer and Sweder van Wijnbergen. Sukuk defaults: on distress resolution in islamic finance. *Qualitative Research in Financial Markets*, 17(2):292 – 311, 2025. doi: 10.1108/QRFM-08-2023-0203.

Muhammad Nooraiman Zailani, Nurul Huda Mohd Satar, Roza Hazli Zakaria, and Mohd Khairi Ismail. Measuring the poverty level of poor and destitute asnaf in kuala lumpur: Application of an enhanced malaysian multidimensional poverty index. *International Journal of Islamic Finance and Sustainable Development*, 17(3):4 – 21, 2025. doi: 10.55188/ijifsd.v17i3.890.

Nor Syahirah Zain and Zulkarnain Muhamad Sori. An exploratory study on musharakah sri sukuk for the development of waqf properties/assets in malaysia. *Qualitative Research in Financial Markets*, 12(3):301 – 314, 2020. doi: 10.1108/QRFM-09-2018-0099.

Mohamad Zaharuddin Zakaria, Ahmad Zaki Salleh, Mushaddad Hasbullah, Abdul Manan Ismail, and Mohd Afiq Ab Jalil. Modelling waqf-based takaful fund as financial aid for the b40 group in malaysia. *Al-Shajarah*, 2019(Special Issue Sharia hand Law):149 – 167, 2019.

Nurudeen Abubakar Zauro, Ram Al Jaffri Saad, and Norfaiezah Sawandi. Enhancing socio-economic justice and financial inclusion in nigeria: The role of zakat, sadaqah and qardhul hassan. *Journal of Islamic Accounting and Business Research*, 11(3):555 – 572, 2020. doi: 10.1108/JIABR-11-2016-0134.

Zawawi, Yuli Yasin, Muhammad Irfan Helmy, Ali Ma'yuf, and Agus Arwani. Waqf and sustainable development law: models of waqf institutions in the kingdom of saudi arabia and indonesia. *Ijtihad: Jurnal Wacana Hukum Islam dan Kemanusiaan*, 23(1):93 – 114, 2023. doi: 10.18326/IJTIHAD.V23I1.93-114.